THE CUSTOMER LOYALTY BLUEPRINT

Customer Shopping Behavior Analysis

Project Overview

Customer Loyalty Blueprint is an end-to-end customer analytics project designed to uncover purchasing behavior, customer loyalty patterns, revenue-driving factors, and product performance trends using SQL, Python, and Power BI.

The project focuses on transforming raw retail transaction data into actionable business insights that can support strategic decision-making in customer retention, personalized marketing, and revenue optimization.

Using a combination of data cleaning, feature engineering, SQL-based exploratory analysis, and interactive Power BI dashboards, the project demonstrates how data analytics can help businesses better understand customer behavior and improve long-term customer engagement.

Problem Statement

Modern retail businesses generate large volumes of customer transaction data every day. However, many organizations struggle to convert this data into meaningful insights that can improve customer retention and business growth.

The primary challenge addressed in this project is understanding:

- Which customer groups contribute the highest revenue
- Which products perform best across categories
- Whether loyalty and subscription programs influence spending behavior
- How discounts affect purchasing decisions
- Which customer segments drive repeat purchases
- How seasonal and demographic trends impact sales performance

Without proper analysis, businesses may invest marketing resources inefficiently, fail to retain valuable customers, or overlook high-performing products and customer segments.

This project aims to solve these challenges by building a structured customer analytics workflow that supports data-driven decision-making.

Dataset Summary

- Rows: 3,900
- Columns: 18

Key Features:

The dataset contains customer shopping behavior data including:

- Customer demographics
- Product categories
- Purchase amounts
- Shipping preferences
- Subscription status
- Payment methods
- Product reviews
- Purchase history
- Seasonal buying trends

Column Name	Description
customer_id	Unique customer identifier
age	Customer age
gender	Gender of customer
item_purchased	Product purchased
category	Product category
purchase_amount	Amount spent
review_rating	Product review score
subscription_status	Subscription membership status
previous_purchases	Number of previous purchases
season	Purchase season
payment_method	Payment mode
shipping_type	Delivery method

Missing Data:

- 37 values in Review Rating column

Data Preparation Steps

- Checked for missing values
- Standardized categorical columns
- Verified data types
- Created age group classifications
- Created purchase frequency metrics
- Prepared the dataset for SQL analysis and dashboard reporting

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Technologies Used

Technology	Purpose
Python	Data Cleaning & Feature Engineering
Pandas	Data Manipulation
MySQL	Data Storage & SQL Analysis
SQL	Business Querying & Insights
Power BI	Interactive Dashboard & Visualization
Jupyter Notebook	Analysis Environment

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

Feature Engineering:

- Created `age_group` column by binning customer ages.
- Created `purchase_frequency_days` column from purchase data.

Data Consistency Check:

- Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

Database Integration:

- Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using SQL (Business Transactions)

Key Analytical Questions

Revenue Analysis

- What is the total revenue generated by male vs female customers?
- Which age groups contribute the highest revenue?
- Which categories generate the highest revenue?

gender	total_revenue
Male	157890
Female	75191

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

category	rev_per_customer	total_rev
Footwear	60.26	36093
Clothing	60.03	104264
Accessories	59.84	74200
Outerwear	57.17	18524

Customer Loyalty Analysis

- Do subscribed customers spend more?
- What percentage of customers are repeat buyers?
- Which customers are classified as loyal customers?

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.4919	62645
No	2847	59.8651	170436

repeat_custperc
97.87

customer_segment	Num_of_customers
Loyal	3116
Returning	701
New	83

Product Performance Analysis

- Which are the top purchased products within each category?
- Which products have the highest review ratings?
- Which products rely heavily on discounts to sell?

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

item_purchased	avg_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer Segmentation

- Segment customers into New, Returning, and Loyal categories
- Identify high-value customers based on total spending

customer_segment	Num_of_customers
Loyal	3116
Returning	701
New	83

customer_id	total_spending	customer_segmt
43	100	High Value
96	100	High Value
194	100	High Value
205	100	High Value
244	100	High Value
249	100	High Value
456	100	High Value
519	100	High Value
582	100	High Value
616	100	High Value
770	100	High Value
862	100	High Value
1209	100	High Value
1301	100	High Value
1406	100	High Value
1413	100	High Value

Seasonal & Behavioral Analysis

- Which season generates the highest sales?

category	season	total_sales
Accessories	Fall	19874
Clothing	Spring	27692
Footwear	Spring	9555
Outerwear	Fall	5259

Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

Dashboard Features

KPI Tracking

- Number of Customers
- Average Purchase Amount
- Average Review Rating
- Subscription Status Analysis

Interactive Filters

Users can dynamically filter insights by:

- Subscription Status
- Gender
- Category

- Shipping Type

Visualizations Included

- Revenue by Category
- Sales by Category
- Revenue by Age Group
- Sales by Age Group
- Customer Distribution by Subscription Status

- Interactive Customer Filters

Key Insights Generated

1. Revenue Contribution by Demographics

Young adult customers contributed a significant portion of total revenue, indicating strong purchasing activity within this segment.

2. Subscription Behavior

Subscribed customers demonstrated stronger repeat purchase behavior and higher engagement.

3. Product Category Performance

Certain product categories consistently generated higher revenue, indicating stronger customer demand.

4. Discount Dependency

Some products showed a high dependency on discounts, suggesting that promotional pricing strongly influenced customer purchases.

5. Customer Loyalty Trends

Repeat buyers represented a significant share of the customer base, highlighting the importance of customer retention strategies.

6. Category Performance

Clothing generated the highest sales and revenue contribution compared to other categories, indicating stronger customer demand within this segment.

Business Impact

This project demonstrates how customer analytics can directly contribute to business growth and strategic planning.

Potential Business Benefits

1. Improved Customer Retention

By identifying loyal and repeat customers, businesses can design targeted retention campaigns and personalized offers.

2. Better Marketing Decisions

Analyzing discount dependency and customer segmentation helps marketing teams optimize promotional strategies.

3. Revenue Optimization

Understanding high-performing categories and products allows businesses to allocate inventory and advertising budgets more effectively.

4. Enhanced Customer Experience

Behavioral insights can support personalized recommendations and loyalty programs that improve customer satisfaction.

5. Data-Driven Decision Making

Interactive dashboards enable stakeholders to monitor key business metrics and make informed operational decisions.

Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs:** Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy:** Balance sales boosts with margin control.
- **Product Positioning:** Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups and express-shipping users.

Learning Outcomes

This project strengthened practical skills in:

- SQL querying and window functions
- Data cleaning using Python
- Business intelligence and dashboard design
- Customer segmentation techniques
- KPI analysis and storytelling
- Translating business problems into analytical solutions

Conclusion

Customer Loyalty Blueprint demonstrates how customer transaction data can be transformed into meaningful business intelligence using SQL, Python, and Power BI.

The project combines data engineering, business analysis, and visualization techniques to uncover customer behavior patterns, identify revenue opportunities, and support data-driven decision-making.

Through customer segmentation, loyalty analysis, product performance evaluation, and dashboard reporting, the project highlights the value of analytics in improving customer retention, optimizing marketing strategies, and enhancing overall business performance.